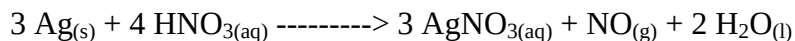
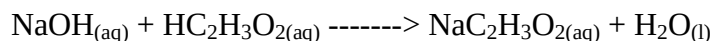


Stoichiometry Involving Solutions Worksheet

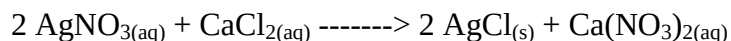
1. Calculate the number of mL of 2.00 M HNO₃ solution required to react with 216 grams of Ag according to the equation.



2. Calculate in mL the volume of 0.500 M NaOH required to react with 3.0 grams of acetic acid. The equation is:

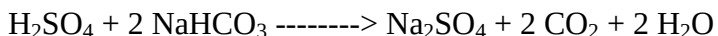


3. Calculate the number of grams of AgCl formed when 0.200 L of 0.200 M AgNO₃ reacts with an excess of CaCl₂. The equation is:



4. Calculate the mass of AgCl formed when an excess of 0.100 M solution of NaCl is added to 0.100 L of 0.200 M AgNO₃.

5. To neutralize the acid in 10.0 mL of 18.0 M H₂SO₄ that was accidentally spilled on a laboratory bench top, solid sodium bicarbonate was used. The container of sodium bicarbonate was known to weigh 155.0 g before this use and out of curiosity its mass was measured as 144.5 g afterwards. The reaction that neutralizes sulphuric acid this way is as follows.



Was sufficient sodium bicarbonate used? Calculate the limiting reactant and the maximum yield in grams of sodium sulphate.